

# Generate Equivalent Expressions

Lesson 6-8

**Name:**

**Date:**

**Class:**

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## TODAY'S OBJECTIVES

**Content Objective:** I can use the commutative, associative, and identity properties to rewrite expressions.

**Language Objective:** I can explain my reasoning using the words commutative property, associative property, and identity property.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Properties of Operations

Commutative Property

Associative Property

Identity Property

Property

Distributive Property



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Rules that let you reorder or regroup without changing the

2

The rule that order does not change a sum or product, like  $a + b = b + a$ , is the

3

The rule that grouping does not change a sum or product, like  $(a + b) + c = a + (b + c)$ , is the

4

The rule that adding 0 or multiplying by 1 keeps a number the same is the

5

A rule that is always true in math is a .

6

The rule that lets you multiply a number across a sum, like  $a(b + c) = a \times b + a \times c$ , is the .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**What properties did the engineer use, and why does rearranging make the calculation easier?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Properties of Operations** — Rules that let you reorder or regroup numbers without changing an expression's value.  
*Propiedades de las operaciones — Reglas que permiten reordenar o reagrupar números sin cambiar el valor de una expresión.*

☐ **Commutative Property** — You can change the order and get the same answer.  
*Propiedad conmutativa — Puedes cambiar el orden y obtener la misma respuesta.*

☐ **Associative Property** — You can change the grouping and get the same answer.  
*Propiedad asociativa — Puedes cambiar la agrupación y obtener la misma respuesta.*

☐ **Identity Property** — Adding 0 or multiplying by 1 keeps the same value.  
*Propiedad de identidad — Sumar 0 o multiplicar por 1 mantiene el mismo valor.*

☐ **Property** — A rule that is always true in math.  
*Propiedad — Una regla que siempre es verdadera en matemáticas.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The engineer used the \_\_\_\_ property to change the \_\_\_\_ and the \_\_\_\_ property to change the \_\_\_\_ . This makes it easier because  $4 \times 25 =$  \_\_\_\_, which is a friendly number to multiply by.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The commutative property lets you change the order of a sum without changing the total.

**Evidence:** Students name the commutative property and explain that  $3 + 5 + 2$  gives 10 in any order because reordering addends does not change the sum.

**Reasoning:** This evidence connects the definition of properties of operations to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The engineer used the \_\_\_\_ property to change the \_\_\_\_ and the \_\_\_\_ property to change the \_\_\_\_\_. This makes it easier because  $4 \times 25 = \underline{\hspace{1cm}}$ , which is a friendly number to multiply by.
- My answer is \_\_\_\_\_.

*Mi respuesta es \_\_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_\_.  
*Mi afirmación es \_\_\_\_\_.*
- I know because \_\_\_\_\_.  
*Lo sé porque \_\_\_\_\_.*
- So \_\_\_\_\_.  
*Entonces \_\_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
  - **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
  - **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*
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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

